Getting Started

Laptop/bwUniCluster 3.0

Julia on Your Laptop

Prerequisites

Course materials

https://github.com/carstenbauer/JuliaKIT26

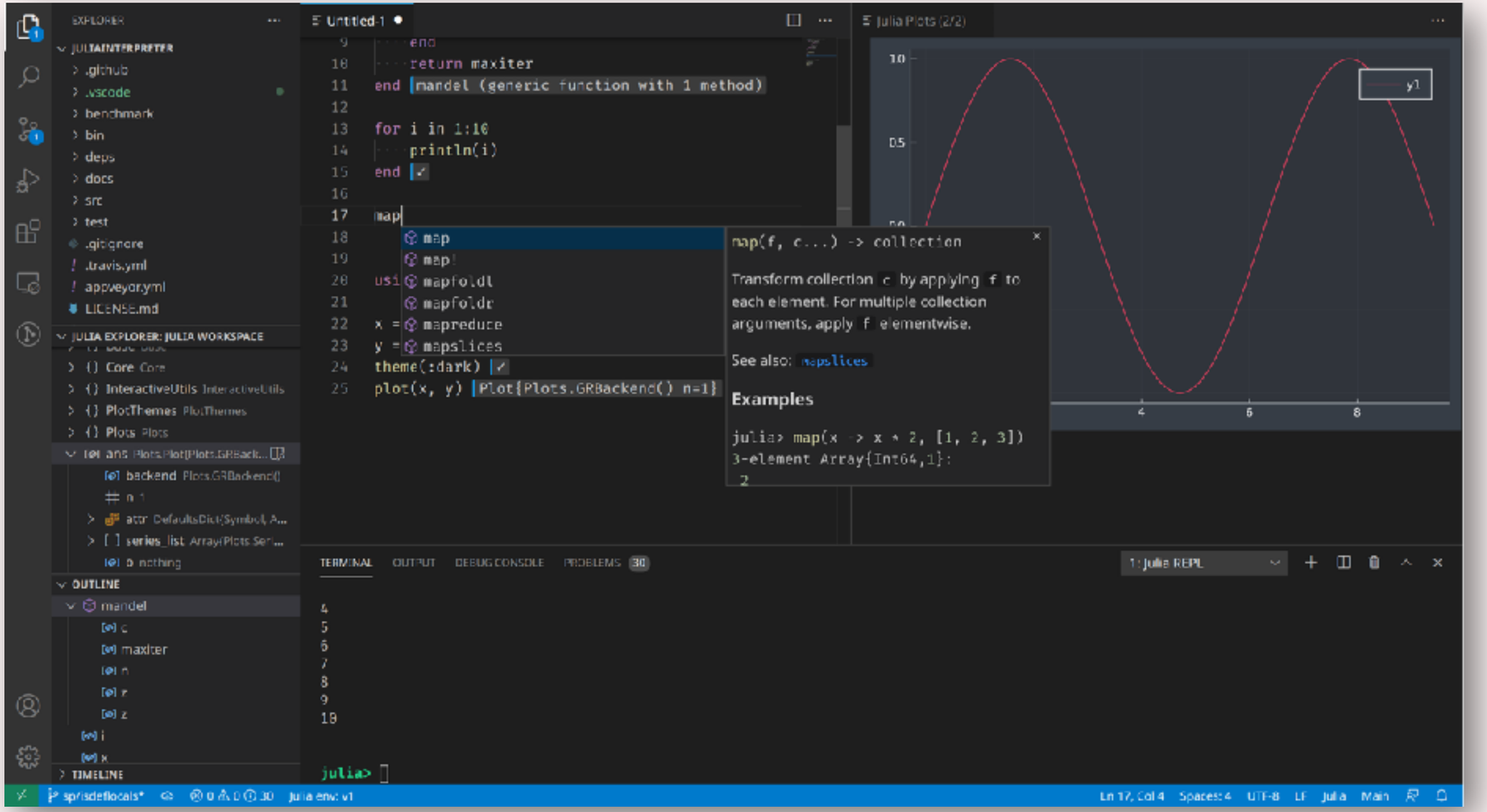
Julia

```
julia (version 1.12)
```

Jupyter + VS Code

```
jupyter lab
code
```

Julia VS Code integration via extension.



Julia on bwUniCluster 3.0

Access bwUniCluster 3.0 via SSH

Terminal

```
ssh <yourlogin>@uc3.scc.kit.edu
```

VS Code

Remote-SSH: Connect to Host…

```
<yourlogin>@uc3-login1.scc.kit.edu
<yourlogin>@uc3-login2.scc.kit.edu
```

Side remark: Accessing compute nodes with VS Code

On the target node

```
module load code
code tunnel
```

On your laptop

In VS Code:

`Remote Tunnels: Connect to Tunnel`

Loading software modules

For this course

```
module load juliahpc
```

This loads Julia, CUDA and OpenMPI.

Outside of the course: If there is no (working) system module, use standard binaries provided by `juliaup`.

Comment: Julia depot is on the parallel file system.

`JULIA_DEPOT_PATH` = where Julia stores stuff

- packages
- binary dependencies
- ...

Why (typically) not `$HOME`?

- Quotas
- Can be read-only for compute jobs

Comment: Julia VSCode extension requires a wrapper.

```
[...]

# Load modules
module load juliahpc

# Pass on all arguments
exec julia "${@}"
```

Jobs are scheduled with SLURM.

Submit a job:

```
sbatch job_script.sh
```

Check on your queued/running jobs:

```
squeue --me
```

Let's try it!

→ cd JuliaKIT26

→ git pull

→ jupyter lab

→ exercises/Day1/1_onboarding